Statistical Model Checking of Ethical Autonomous System Under Uncertainty

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Abstract

Artificial intelligence based systems are omnipresent in our daily lives. It is imperative that the decision making processes in these systems are ethical. The evaluation of ethical decision making becomes challenging when system models are constructed with imprecise data and often, do not address environmental fluctuations during their operations. A tractable formalism, based on interval discrete-time Markov chain to incorporate uncertainty, is constructed. Deontic logic based ethical rules are encoded in the formalism. Statistical model checking is performed on a prototype to demonstrate the computational feasibility and address epistemic uncertainty in the model. Temporal logic queries is posed on the model abstraction and the effectiveness of the results are discussed.

Introduction

Artificial intelligence (AI) based autonomous systems are becoming omnipresent. Ethical decision making is critical in eliminating harm caused by AI technology. The main theme for use of AI technologies is to provide an efficient tool for performing tasks while eliminating any harm caused to public or create obstacles in smooth functioning of activities of daily lives. Therefore, ethical decision making becomes critical in the implementation of AI technologies. The evaluation of system properties of an AI system, based on correct ethical decision making can be performed using formal verification.

Formal verification is an established research area that has effective tools for hardware and software verification. Application of formal verification techniques, in particular model checking (Clarke, Emerson, and Sistla 1986) has been effective in verification of software and hardware. The model, a finite state machine is constructed and properties, stated as specification, are represented in the form of logic formula. The validity of the specification is evaluated in the model. If the specification is valid, it implies that the property holds true in the model else, there is a counterexample generated. The challenge in model checking is the state explosion problem. Large problem sizes are computationally intensive. The challenges are increased when model checking is performed for uncertain stochastic structures such as interval discrete-time Markov chain (IDTMC). A variant of model checking, statistical model checking is scalable in comparison to model checking. Therefore, statistical model checking performed on interval discrete-time Markov chain is computationally feasible for evaluation of the correctness of a system.

Formal verification for evaluation of ethical properties in a system is critical. However, the ethical actions defined as ethical properties are dependent on the environment for an autonomous system. The probabilities for actions change under environmental fluctuations. The actions are categorized using Deontic logic rules such as obligations, permissible actions and forbidden actions. The goal of this work is to evaluate ethical decision making of a autonomous system. The contribution of this work is the following:

1. A Deontic logic-based description of a model represented by interval discrete-time Markov chain. The interval discrete-time Markov chain addresses the uncertainty in the model, and an approximation is constructed for tractable model checking.
2. Statistical model checking on the tractable formalism using Deontic logic is performed on a prototype of an aircraft in an airport. Bayesian model averaging is evaluated on the statistical inference of the output of the experiments that are conducted.

The tractable formalism on IDTMC combined with statistical model checking and statistical inference presents a way to handle computational efficiency and accuracy under uncertainty.

Background

The background of this work is based on the foundations of statistical model checking on interval discrete-time Markov chain, statistical inference and

Statistical Model Checking on Discrete Time Markov Chain: Statistical model checking (Younes and Simmons 2002, 2006) (SMC) have been used in the verification of Cyber-Physical systems (Clarke and Zuliani 2011) and systems biology (Zuliani 2015). Details on statistical model checking have been reported (Agha and Palmskog 2018; Legay et al. 2019). The challenge in verification is the uncertainty due to environmental fluctuations and loss of details as a result of discretizations of continuous parameters. SMC comes with many advantages, including the ability to analyze large state spaces, and support systems whose implementation systems

are unknown and cannot be simulated on demand. Statistical tests can be used to reject proposed stochastic models of systems, such as DTMCs. One way is by checking whether the key SMC requirement of stochastic behavior is satisfied.

Model Checking on IDTMC: Uncertainty in stochastic systems is represented by interval discrete-time Markov chain. Model checking of IDTMC was reported (Baier et al. 2005). A theoretical formalism of using probabilistic computation tree logic (PCTL) specification on models represented by interval discrete-time Markov chains were constructed (Sen, Viswanathan, and Agha 2006). Recent studies have used simulation based approaches for verification of continuous-time Markov chain with uncertain rates of transition (Badings et al. 2022). The challenges of simulation based methods are errors if the simulations is not proximal to the values of the correct rates of transition.

Ethical Artificial Systems: The evaluation of ethics in AI systems is critical for safety. The relationships between AI systems and reasoning tools for evaluation of ethics in a system have been studied(Bonnemains, Saurel, and Tessier 2018). Formal verification of ethical property in multi-agent systems is evaluated under conflicts in moral rules (Mermet and Simon 2016). Ethical decision making for human-agent behavior is studies on a prototype representing unmanned aerial vehicle (Abeywickrama, Cirstea, and Ramchurn 2019). An investigation using formal verification techniques for selection of ethical choices for autonomous systems has been performed (Dennis et al. 2016). Reasoning using logic on ethics using Deontic logic (Horty 2001) has been reported (Hilpinen and McNamara 2013). Model checking of ethical rules under uncertainty using IDTMC is investigated (Godawa and Ghosh 2025). However, this work differs the published work (Godawa and Ghosh 2025) by mitigating the explosion in the number of submodels, applying model averaging for efficiency and performing statistical model checking on the formalism. In this work, statistical model checking was performed. Prior work (Godawa and Ghosh 2025), had demonstrated probabilistic model checking.

Preliminaries

Definition 1 *Discrete Time Markov Chains A discrete-time Markov chain (DTMC) is a tuple:* $\mathcal{K}_m\langle S, S_0, \iota_{init}, P, L\rangle$ *where:*

- S *is a finite set of states.*
- S_0 *is the set of initial states.*
- $P : S \times S \to [0,1]$ *, where* P *represents the probability matrix and* $\sum_{s,s' \in S} \mathcal{P}(s,s') = 1.$
- $\iota_{init} : S \to [0,1]$ *where* $\sum_{s \in S} \iota_{init}(s) = 1$ *is the initial distribution.*
- $L : S \to 2^{AP}$*, where* L *is a labeling function and* AP *the set of atomic propositions.*

Definition 2 *(Interval Discrete Time Markov Chain (IDTMC) (Benedikt, Lenhardt, and Worrell 2013)) is a tuple* $\mathcal{K}_I = \langle S, \iota_{init}, M_l, M_u\rangle$ *where*

- S *is a finite set of states.*
- $K_l, K_u : S \times S \to [0,1]$ *with* $K_l \leq K_u$*,* K_l *and* K_u *are transition matrices.*
- $\iota_{init} : S \to [0,1]$ *where* $\sum_{s \in S} \iota_{init}(s) = 1$ *is the initial distribution.*

Here, ϕ represents a temporal logic formula that represents the properties of a system. The model, $\mathcal{K}$ represents the description of the system. The system description when represented by a stochastic structure, $\mathcal{K}_s$, then the model checking is performed on a stochastic structure and the validity of the formula is quantified by probability,p.
Details on probabilistic temporal logic can be found in the literature (Baier and Katoen 2008; Aziz et al. 1996; Hansson and Jonsson 1994). Probabilistic model checking is performed on stochastic structures such as discrete-time Markov chain, Markov decision process and continuous-time Markov chain. In this work, the logics that will be used to represent the specifications are PCTL.

Probabilistic Computation Tree Logic

We describe the syntax of PCTL and semantics of PCTL over DTMCs and MDP. The following is summarization from the published dissertation (Parker 2003).

Syntax of PCTL:

$$\phi ::= true \mid p \mid \phi \wedge \phi \mid \neg\phi \mid \mathcal{P}_{\oplus\mathcal{J}}[\psi]$$

$$\psi ::= X\phi \mid \phi\mathcal{U}^{\leq k}\phi \mid \phi\mathcal{U}\phi$$

where p is an atomic proposition, $\oplus \in \{\leq, <, \geq, >\}, \mathcal{J} \in [0,1]$ and $k \in N$. ϕ, ψ are state and path formula respectively. ϕ and ψ are state and path formulas respectively. Each of these formulas are interpreted over a DTMC. Each state of DTMC is labeled from the set of atomic proposition. Specification is represented in the form of a state formula. Path formula ψ are preceded by the probability path operator $\mathcal{P}$.DTMC satisfies $\mathcal{P}_{\oplus\mathcal{J}}$ is the probability of a path from s satisfying ψ is in the bound stated by $\oplus p$. The path forumla,$X\phi$ is true if ϕ is satisfied in the next state. The formula $\phi_1\mathcal{U}^{\leq k}\phi_2$ is true if ϕ_2 is satisfied within k time-steps and ϕ_1 is true at that point. Similar is the description of $\phi_1\mathcal{U}\phi_2$ where ϕ_2 is true some point in future till then ϕ_1 is true. We refer the reader for a details of PCTL(Parker 2003).

Statistical Preliminaries

Approximation of Probability Distribution: The probability distribution are approximated by the following algorithms- Expectation-maximization (EM) and Markov chain Monte Carlo (MCMC). The EM algorithm constructs the probability distribution using maximizing the likelihood function. However, the MCMC is based on simulations by drawing samples and constructs an approximate probability distribution.

Bayesian model averaging (BMA): learns parameters for all candidate models, and then combine the estimates according to the posterior probabilities of the associated models. The BMA estimate, $P(t)$, provides and assessment of the result t that takes in account all scenarios simultaneously.

Model of Deontic Logic

Deontic logic has components- obligations, permissible and forbidden actions. For a set of actions, $\mathcal{A}$, each action $a \in \mathcal{A}$ can be labeled as obligation, permissible or forbidden action. Notation: O_a, P_a and Fr_a denote an action,a as obligation, permissible and forbidden. Additionally, a forbidden action, $F_\emptyset$ represents an action that does not have the properties of forbidden action. For clarity, F_a will denote any forbidden action, including $F_\emptyset$.

A triple of actions (t-action), $\langle O_{a_m}, P_{a_n}, F_{a_p} \rangle$ where $m, n, p \in N$ represents any combination of obligation, permissible and forbidden actions.

An *outcome* is denoted by, $s \rightarrow s'$ where s, s' are t-actions. Notation: $s = \langle O_{a_m}, P_{a_n}, F_{a_p} \rangle$ and $s' = \langle O_{a_{m'}}, P_{a_n{}'}, F_{a_{p'}} \rangle$ and at least one of the following is true $O_{a_m} \neq O_{a_{m'}}$ or $P_{a_n} \neq P_{a_{n'}}$ or $F_{a_p} \neq F_{a_{p'}}$. The reading of an outcome, $s \rightarrow s'$ is after an completion of one of the actions in s leads to the execution of actions in s'. We define types of outcome with specific properties:

1. Admissible Outcome (a-outcome): An outcome,$s \rightarrow s'$ is admissible if for t-actions, $s = \langle O_{a_m}, P_{a_n}, F_{a_p} \rangle$ and $s' = \langle O_{a_{m'}}, P_{a_n{}'}, F_{a_{p'}} \rangle$, there is at least one of the following is true $O_{a_m} \neq O_{a_{m'}}$ or $P_{a_n} \neq P_{a_{n'}}$ or $F_{a_p} \neq F_{a_{p'}}$.
2. Inadmissible Outcome(i-outcome): An outcome where the t-action is $s = \langle O_{a_m}, P_{a_n}, F_{a_p} \rangle$ and $F_{a_p} \neq F_\emptyset$.
3. Total outcome (t-outcome): A outcome, $s \rightarrow s'$ where $s' = s$.

The i-outcome requires the t-actions to have a forbidden action while it is not necessary for t-outcome to have a forbidden action. A trace of t-actions ,π is a sequence of given by $\pi = s_1, s_2, \ldots, s_k$ where $k \in N$.

Model Description

The model abstraction using Deontic logic for ethics decision making is described in the following steps:

1. Construc IDTMC, $\mathcal{K}_I$ representing ethical actions.
2. Construct a set of submodels, K_s from $\mathcal{K}_s$.
3. Construct DTMC $\mathcal{K}_m$ for each $\mathcal{K}_s \in K_s$.
4. Perform model averaging based on the random numbers of the intervals so that the sum of the random numbers is equal to one.
5. Perform statistical model checking using PCTL formula on each $\mathcal{K}_m$ such that $\mathcal{K}_m \in K_m$.
6. Construct of probability distribution from outputs of PCTL queries on each $\mathcal{K}_m$ using EM/MCMC algorithms.
7. Perform statistical inference on the constructed probability distribution..

Fig 1 shows the schematic diagram of the steps.

In the first step of model abstraction, the ethical decisions are constructed using IDTMC, $\mathcal{K}_I$ and then, submodels,$\mathcal{K}_s$ are constructed using the partition of the intervals into subintervals, In the second step, a random number is selected from each subinterval such that the sum of random numbers extracted is equal to 1. The set of random numbers fulfilling the summation equal to 1, is the probability distribution of a dtmc,$\mathcal{K}_m$. The set of dtmcs constructed from the set of submodels, K_s is denoted by K_m.

Construction of IDTMC representing actions: Given (1) a set of actions, $\mathcal{A}$, (2) for each action, $a \in \mathcal{A}$ is labeled as obligations (O), permission(P) and forbidden(F) actions, (3) a set of t-actions, $\mathcal{T}_a$ and (4) a set of outcomes, Out The IDTMC, $\mathcal{K}_I = \langle S, \iota_{init}, K_l, K_u \rangle$ is described as follows:

1. AP is the set of all atomic formulas of the form,$s = \langle O_a, P_a, F_a \rangle$ where $\langle O_a, P_a, F_a \rangle \in \mathcal{T}_a$.
2. S is the set of all subsets s of AP where exactly one of the formula, $\langle O_a, P_a, F_a \rangle$ is in s.
3. S_0 is the set of initial states of $\mathcal{K}_m$. An initial state contains the t-actions where the permissible action is to be the initial action. Therefore,
4. The label(edge label) on a transition is the outcome, $out \in Out$. The labeled transition is represented by a triple,$\langle s, e, s' \rangle$ where $e = out$ and $out \in \{a - outcome, i - outcome, t - outcome\}$
5. The edge labels are probability intervals of the form ,$[p_l, o_u]$ where $p_l.p_u$ are the lower and upper bounds of the interval, and are values in the transition matrices, K_l and K_u, respectively.

Initially, $\mathcal{K}_I$ is constructed wit a transition for each state to every other state. The edge labels with $a - outcome$ remain in the structure. The transition with edge label, $i - outcome$ are pruned. The transitions with edge label, $t - outcome$ are pruned and a self-loop will be constructed on the outgoing state of the outcome. After pruning and adding transitions to $\mathcal{K}_I$, submodels are generated.

Construction of submodels from $\mathcal{K}_I$: The tractable abstraction from the IDTMC is constructed by discretizing the intervals into subintervals. The subintervals are constructed from the user-defined value. If the number of subinterals is two, then the subintervals are $(a, \frac{a+b}{2}], (\frac{a+b}{2}, b)$ from an interval,(a, b). The subintervals are in the feasbile space such that there exits a number from each subinterval whose sum is 1. Each subintervals form submodels. For each submodel, a random values from the subintervals are generated such that the sum is equal to one. If the sum is not equal to 1, then the random values are rejected. The computations are performed and a set of submodels are generated from the cross products of subintervals at different lengths of the IDTMC.

Statistical Inference on the Model: The output from each PCTL formula when posed on each dtmc, $\mathcal{M}_m \in M_m$ are recorded. The output values are used for construction of probability distribution using EM and MCMC algorithm. The confidence interval are computed on the constructed probability distribution of each of PCTL query.

Simulation

The proposed formalism for evaluating ethical properties of a system was experimented on a published case study (Ghosh and Smith 2025). Fig 2 demonstrates the movement

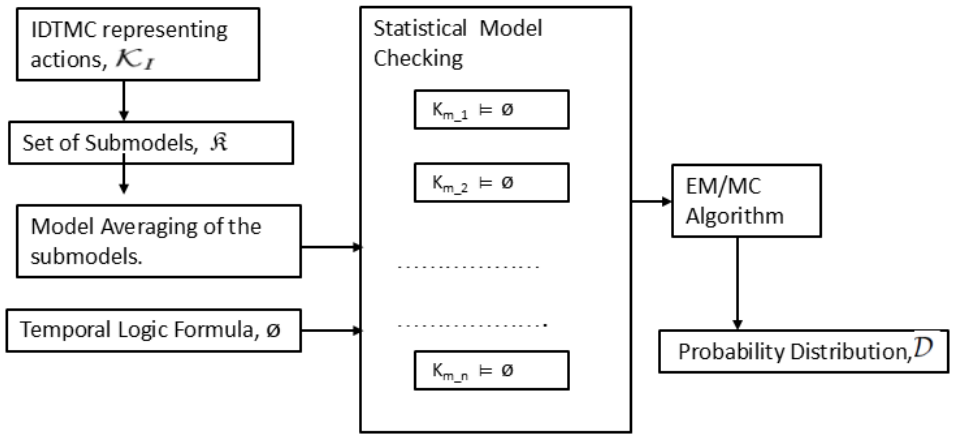


Figure 1: Schematic diagram of statistical inference of IDTMC, $\mathcal{K}_I$ representing ethical actions.

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of the aircraft in an airport. The motion of the aircraft is confirm that the aircraft does not collide with any obstacle. The collision with any of the obstacles imply that the aircraft will stay on that state labeled with obstacles (represented with a self loop). The state, labeled with take_off represents that the aircraft will take_off and will stay in that state. The data and scripts are available online (https://zenodo.org/records/21481790). The PCTL queries were executed in PRISM model checker (Kwiatkowska, Norman, and Parker 2002) for statistical model checking.

Construction of the intervals-skewed intervals

The experiments were conducted on Apple M5 chip, 16 GB memory, 10 cores, running MacOS 26.2. The queries were completed within 1 second. The sample set of queries that were evaluated are:

Q1. Probability of reaching the goal state? P=? [F goal_1].

Q2. Probability of hitting an obstacle? P=? [F stop_1].

Q3. Probability of reaching take_off P=? [F goal_1].

The subintervals are desribed in Table 1. The results from the construction of DTMCs from IDTMC is demonstrated in Table 3. Experiments were carried out using PCTL queries. The discrete-time Markov chain with probability 0.5 is represented by the intervals. Table 3 shows the construction of the probabilities from $\mathcal{K}_I$ was transformed to dtmcs. The statistical significance of 95% confidence interval was computed. Additionally, the width of the intervals are 0.1. Fig 2 shows the results of the queries, Q1, Q2 and Q3 for each subinterval. Fig 4 shows distribution for submodels with interval size 0.1 for each query using EM algorithm. Fig 5 show the probability distributions for query 1, query 2 and query 3 using MCMC algorithm, respectively. The probability distributions for each of the queries follows a normal distribution.

Construction of model averages

As detailed in the previous section, we construct intervals using a set of 5 subintervals. Model averages were performed for each of the 4 intervals by taking the average of the 5 p_1, p_1' values, as seen in Table 3. This yields 4 intervals, which the 3 PCTL formulas are simulated on.

Bayesian Model Averaging describes how an observed data update prior beliefs for θ to posterior beliefs and can be used for models to compute the posterior model probability (PMP).

The number of submodels constructed were directly proportional to the number of subintervals and potentially, could have been the cartesian products of the subintervals. The subintervals were constructed from intervals whose midpoint was not the same as the probabilities in the discrete-time Markov chain, represented in Fig 2. The imprecision was added by not creating the intervals with midpoint. The methodology has its limitations.

1. Model averaging created the formalism computationally tractable as the probabilities of the submodels were averaged and not used as different submodels.
2. MCMC and EM are evaluated on the formalism for approximation of the probability distributions. Table 4 demonstrates the different from the constructed value and the original probabilities is better in the case of EM algorithm.

Conclusion

The work demonstrates how ethical decision can be evaluated under uncertainty using Deontic logic rules and statistical model checking on an interval discrete-time Markov chain. The reasoning mechanism is tractable and provides statistical guarantee of the temporal logic queries for statistical model checking. The experiments on the published case study demonstrate promising results. However, the number

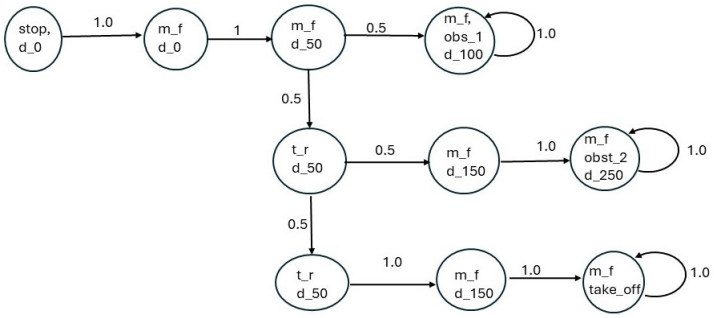


Figure 2: Movement of Aircraft in Airport (Ghosh and Smith 2025).Notation: m_f and t_r imply the actions, moving forward and turning to the right, respectively. The distances moved by units,0,50,100,150 are represented by d_0,d_50,d_100 and d_150.

	S	E	S'	E'	n'	SI
1	0.48	0.58	0.4	0.6	10	{[0.48, 0.5], [0.5, 0.52], [0.52, 0.54], [0.54, 0.56], [0.56, 0.58]}
2	0.47	0.53	-	-	-	{[0.47, 0.482], [0.482, 0.494], [0.494, 0.506], [0.506, 0.518], [0.518, 0.53]}
3	0.43	0.53	0.43	0.57	7	{[0.43, 0.45], [0.45, 0.47], [0.47, 0.49], [0.49, 0.51], [0.51, 0.53]}
4	0.45	0.55	-	-	-	{[0.45, 0.47], [0.47, 0.49], [0.49, 0.51], [0.51, 0.53], [0.53, 0.55]}

Table 1: Results from the subinterval selection process, where S refers to the start of the skewed interval, E refers to the end of the skewed interval, S' refers to the start of the encompassing interval, if one exists, E' refers to the end of the encompassing interval, n' refers to the total number of subintervals in the encompassing set, and SI is the set of skewed subintervals, where $|SI| = 5$. Note that the lack of a value in a cell indicates that no encompassing set was required to model the skewed set.

	Subinterval	p_1	p_1'	Q1	Q2	Q3
Interval 1	[0.48, 0.5]	0.4975444029379343	0.5024555970620657	0.491	0.513	0.499
	[0.5, 0.52]	0.5099091280164729	0.4900908719835271	0.486	0.535	0.511
	[0.52, 0.54]	0.5322265448034105	0.4677734551965895	0.482	0.504	0.451
	[0.54, 0.56]	0.5452206059359723	0.4547793940640277	0.436	0.566	0.446
	[0.56, 0.58]	0.5750496650797166	0.42495033492028345	0.387	0.584	0.436
Interval 2	[0.47, 0.482]	0.4767619697824629	0.5232380302175371	0.540	0.452	0.541
	[0.482, 0.494]	0.4901625974244958	0.5098374025755041	0.520	0.473	0.526
	[0.494, 0.506]	0.49442173064152456	0.5055782693584754	0.514	0.473	0.513
	[0.506, 0.518]	0.5067106466874152	0.49328935331258483	0.511	0.520	0.503
	[0.518, 0.53]	0.528057128975047	0.471942871024953	0.484	0.542	0.463
Interval 3	[0.43, 0.45]	0.443459029829656	0.556540970170344	0.566	0.436	0.551
	[0.45, 0.47]	0.45943918228976666	0.5405608177102333	0.550	0.470	0.563
	[0.47, 0.49]	0.4784535054490635	0.5215464945509365	0.517	0.497	0.507
	[0.49, 0.51]	0.5057151595798565	0.49428484042014353	0.498	0.508	0.511
	[0.51, 0.53]	0.526961214001842	0.47303878599815796	0.472	0.541	0.461
Interval 4	[0.45, 0.47]	0.4692566129190307	0.5307433870809692	0.528	0.475	0.513
	[0.47, 0.49]	0.4894245616745444	0.5105754383254556	0.501	0.470	0.509
	[0.49, 0.51]	0.4977581356869125	0.5022418643130875	0.492	0.508	0.536
	[0.51, 0.53]	0.5252940422781874	0.47470595772181257	0.478	0.513	0.488
	[0.53, 0.55]	0.5489494404460169	0.4510505595539831	0.453	0.543	0.439

Table 2: Discrete probabilities generated for each subinterval, and the resulting query results from each pair. p_1 and p_1' correspond to the probabilities of selecting the first or second probabilistic branch in the DTMC model, respectively.

of submodels can increase exponentially, an efficient model abstracted will be constructed.

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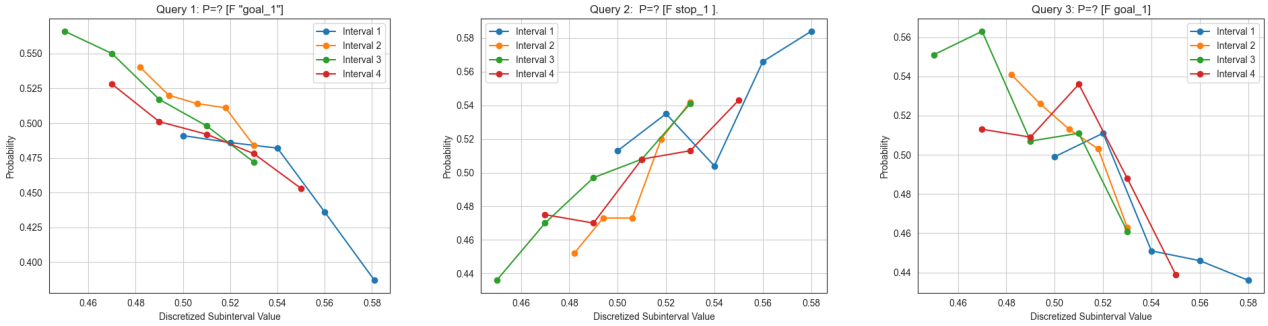
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	Subinterval	p_1	p_1'	Q1	Q2	Q3
Interval 1	[0.45, 0.47]	0.45499164791165475	0.5450083520883452	0.545	0.455	0.543
	[0.47, 0.49]	0.4789645568481132	0.5210354431518868	0.517	0.483	0.527
	[0.49, 0.51]	0.49865406342332363	0.5013459365766764	0.500	0.500	0.500
	[0.51, 0.53]	0.518507741789924	0.481492258210076	0.488	0.512	0.457
	[0.53, 0.55]	0.5385792741050777	0.4614207258949224	0.488	0.551	0.449
Interval 2	[0.45, 0.47]	0.45499164791165475	0.5450083520883452	0.164	0.836	0.173
	[0.47, 0.49]	0.4789645568481132	0.5210354431518868	0.146	0.854	0.139
	[0.49, 0.51]	0.49865406342332363	0.5013459365766764	0.127	0.873	0.135
	[0.51, 0.53]	0.518507741789924	0.481492258210076	0.126	0.913	0.087
	[0.53, 0.55]	0.5385792741050777	0.4614207258949224	0.116	0.884	0.111
Interval 3	[0.45, 0.47]	0.45499164791165475	0.5450083520883452	0.048	0.950	0.050
	[0.47, 0.49]	0.4789645568481132	0.5210354431518868	0.035	0.960	0.040
	[0.49, 0.51]	0.49865406342332363	0.5013459365766764	0.030	0.971	0.029
	[0.51, 0.53]	0.518507741789924	0.481492258210076	0.021	0.974	0.026
	[0.53, 0.55]	0.5385792741050777	0.4614207258949224	0.019	0.972	0.028
Interval 4	[0.45, 0.47]	0.45499164791165475	0.5450083520883452	0.021	0.989	0.011
	[0.47, 0.49]	0.4789645568481132	0.5210354431518868	0.010	0.990	0.008
	[0.49, 0.51]	0.49865406342332363	0.5013459365766764	0.008	0.993	0.007
	[0.51, 0.53]	0.518507741789924	0.481492258210076	0.004	0.996	0.009
	[0.53, 0.55]	0.5385792741050777	0.4614207258949224	0.003	0.995	0.005

Table 3: Discrete probabilities calculated using the mean of 5 random samples from each subinterval, and the resulting query results from each pair. p_1 and p_1' correspond to the probabilities of selecting the first or second probabilistic branch in the DTMC model, respectively.



(a) Q1. (b) Q2. (c) Q3.

Figure 3: Experimental results from statistically verifying queries 1, 2, and 3 on intervals 1, 2, 3, and 4.

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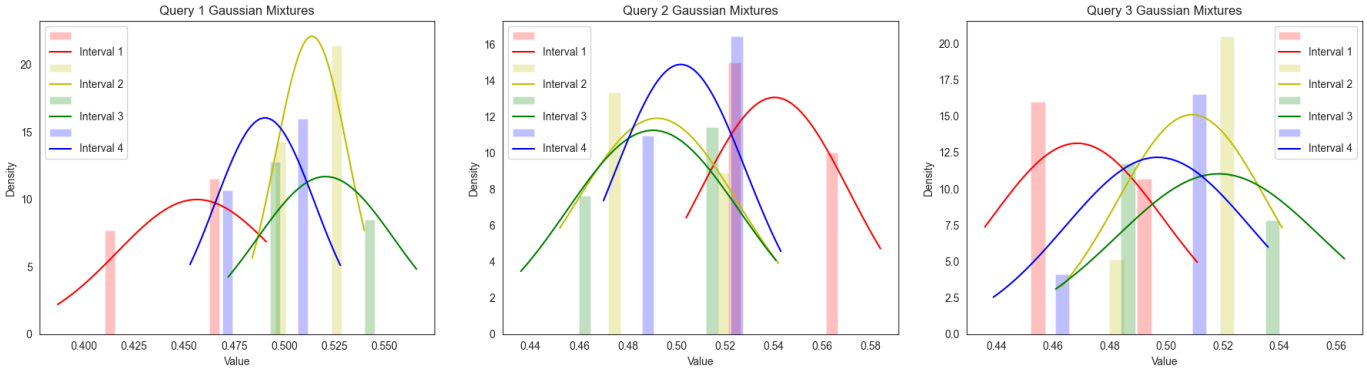
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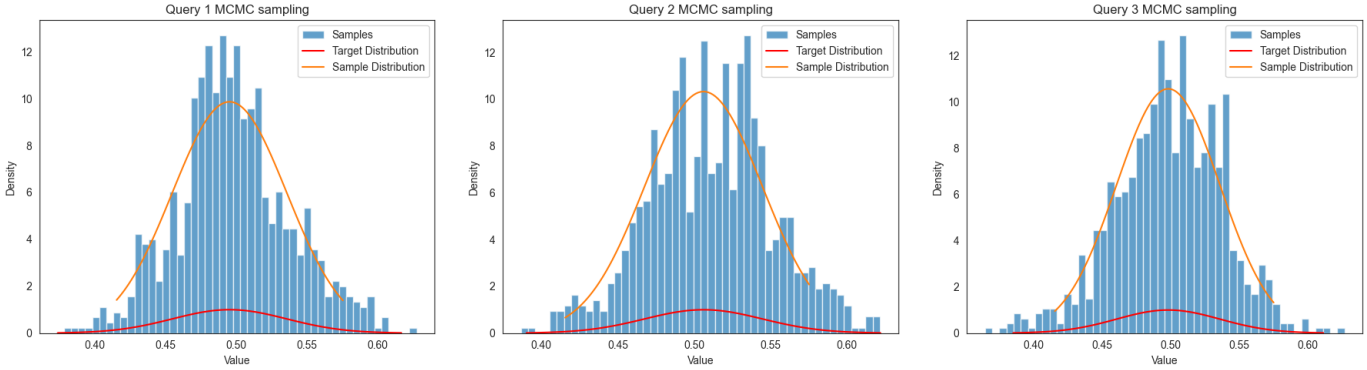
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(a) Q1. (b) Q2. (c) Q3.

Figure 4: EM results for queries 1, 2, and 3 executed on intervals 1, 2, 3, and 4. The interval width is 0.1.



(a) Q1. (b) Q2. (c) Q3.

Figure 5: Results of MCMC sampling performed with statistical data from intervals 1, 2, 3, and 4 for queries 1, 2, and 3. The output of the samples are used to approximate a target posterior distribution.

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Query	Mean	Method	Distribution	SC	Diff
Q1	0.4904	EM	Interval 1	(0.4417, 0.5391)	0.0000
	0.5404	EM	Interval 2	(0.4805, 0.6003)	0.0000
	0.5126	EM	Interval 3	(0.4638, 0.5614)	0.0000
	0.4996	EM	Interval 4	(0.4329, 0.5663)	0.0000
Q2	0.5362	EM	Interval 1	(0.4656, 0.6071)	0.0000
	0.5014	EM	Interval 2	(0.4661, 0.5367)	0.0000
	0.4856	EM	Interval 3	(0.4214, 0.5498)	0.0000
	0.5100	EM	Interval 4	(0.4558, 0.5642)	0.0000
Q3	0.4694	EM	Interval 1	(0.4215, 0.5173)	0.0000
	0.5012	EM	Interval 2	(0.4530, 0.5494)	0.0000
	0.5144	EM	Interval 3	(0.4596, 0.5692)	0.0000
	0.4926	EM	Interval 4	(0.4186, 0.5666)	0.0000
Q1	0.4937	MCMC	Interval 1	(0.4269, 0.5604)	0.0349
	0.4919	MCMC	Interval 2	(0.4225, 0.5612)	0.0085
	0.4918	MCMC	Interval 3	(0.4211, 0.5625)	0.0298
	0.4919	MCMC	Interval 4	(0.4256, 0.5581)	0.0077
Q2	0.5086	MCMC	Interval 1	(0.4371, 0.5802)	0.0276
	0.5073	MCMC	Interval 2	(0.4331, 0.5815)	0.0059
	0.5082	MCMC	Interval 3	(0.4417, 0.5748)	0.0226
	0.5105	MCMC	Interval 4	(0.4381, 0.5829)	0.0005
Q3	0.4957	MCMC	Interval 1	(0.4203, 0.5711)	0.0263
	0.4931	MCMC	Interval 2	(0.4263, 0.5601)	0.0081
	0.4967	MCMC	Interval 3	(0.4272, 0.5663)	0.0177
	0.4951	MCMC	Interval 4	(0.4230, 0.5672))	0.0001

Table 4: Statistical Confidence (SC) Scores for probability distributions constructed from subintervals 1-4 using the EM algorithm and MCMC algorithm on query results for each subinterval. The column, Diff implies the difference between the mean of the probability distribution and the mean of the output values for each interval as seen in Table II..

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